150+ Python Pattern Programs

TOP Python exercises to feed your creativity with if statements and loops



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The coding king

150+ Python Pattern Programs

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Introduction

Develop your creativity by creating beautiful asterisk patterns with if statements and loops using the Python language. this awesome language will allow you to feel more comfortable writing code because you will master the more essential topics of programming by practicing these techniques.

If you are studying computer science at university, this book is perfect for you. They will require you to create these types of patterns that will be very useful to learn all types of advanced techniques and quickly master this wonderful language.

Pattern 1

```
# *****
# *****
# *****
# *****
# *****
```

```
for i in range(1, 6):
    for j in range(1, 6):
        print("*", end="")  # put any character
    print()
```

Pattern 2

```
# *
# **
# ***
# ****
# *****

for i in range(1, 6):
    for j in range(1, i + 1):
        print("*", end="")
    print()
```

Pattern 3

```
# *****
# ****
# ***
# **
# *

for i in range(1, 6):
    for j in range(5, i - 1, -1):
        print("*", end="")
    print()
```

Pattern 4

```
#     *
#    **
#   ***
#  ****
# *****
```

```
n = 5

for i in range(1, n + 1):
    for j in range(n - 1, i - 1, -1):
        print(" ", end="")
    for k in range(1, i + 1):
        print("*", end="")
    print()
```

Pattern 5

```
# *****
#  ****
#   ***
#    **
#     *
```

```
n = 5

for i in range(n, 0, -1):
    for j in range(n - 1, i - 1, -1):
        print(" ", end="")
    for k in range(1, i + 1):
        print("*", end="")
    print()
```

Pattern 6

```
#     *
#    * *
#   * * *
#  * * * *

p_height = 5

for i in range(1, p_height):
    for k in range(p_height - 1, i - 1, -1):
        print(" ", end="")
    for j in range(1, i + 1):
        print("* ", end="")
    print()
```

Pattern 7

```
# * * * * *
#  * * * *
#   * * *
#    * *
#     *
```

```
p_height = 5

for i in range(p_height, 0, -1):
    for k in range(p_height - 1, i - 1, -1):
        print(" ", end="")
    for j in range(i, 0, -1):
        print("* ", end="")
    print()
```

Pattern 8

```
#     *
#    ***
#   *****
#  *******
# *********
```

```
p_height = 5
min_stars = 1

for i in range(p_height):
    for j in range(p_height - 1, i, -1):
        print(" ", end="")
    for k in range(min_stars):
        print("*", end="")
    min_stars += 2
    print()
```

Pattern 9

```
# *********
#  *******
#   *****
#    ***
#     *
```

```
p_height = 5
max_stars = p_height * 2 - 1
p_space = p_height - 1

for i in range(p_height, 0, -1):
    for j in range(p_space, i - 1, -1):
        print(" ", end="")
    for k in range(max_stars):
        print("*", end="")
    max_stars -= 2
    print()
```

Pattern 10

```
# *
# **
# ***
# ****
# ***
# **
# *
```

```
size = 3

for i in range(size, -size-1, -1):
    for j in range(size, abs(i)-1, -1):
        print("*", end="")
    print()
```

Pattern 11

```
#   *
#  **
# ***
#****
# ***
#  **
#   *
```

```
size = 3

for i in range(size, -size-1, -1):
    for j in range(1, abs(i)+1):
        print(" ", end="")
    for k in range(size, abs(i)-1, -1):
        print("*", end="")
    print()
```

Pattern 12

```
# 1****
# 22***
# 333**
# 4444*
# 55555

n = 5

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if j > i:
            print("*", end="")
        else:
            print(i, end="")
    print()
```

Pattern 13

```
# 1****
# 12***
# 123**
# 1234*
# 12345

n = 5

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if j > i:
            print("*", end="")
        else:
            print(j, end="")
    print()
```

Pattern 14

```
# 55555
# 4444*
# 333**
# 22***
# 1****

n = 5

for i in range(n, 0, -1):
    for j in range(1, n + 1):
        if j > i:
            print("*", end="")
        else:
            print(i, end="")
    print()
```

Pattern 15

```
# 12345
# 1234*
# 123**
# 12***
# 1****

n = 5

for i in range(n, 0, -1):
    for j in range(1, n + 1):
        if j > i:
            print("*", end="")
        else:
            print(j, end="")
    print()
```

Pattern 16

```
# ****1
# ***22
# **333
# *4444
# 55555

n = 5

for i in range(1, n + 1):
    for j in range(n, 0, -1):
        if j > i:
            print("*", end="")
        else:
            print(i, end="")
    print()
```

Pattern 17

```
# ****1
# ***21
# **321
# *4321
# 54321

n = 5

for i in range(1, n + 1):
    for j in range(n, 0, -1):
        if j > i:
            print("*", end="")
        else:
            print(j, end="")
    print()
```

Pattern 18

```
# 55555
# *4444
# **333
# ***22
# ****1

n = 5

for i in range(n, 0, -1):
    for j in range(n, 0, -1):
        if j > i:
            print("*", end="")
        else:
            print(i, end="")
    print()
```

Pattern 19

```
# 54321
# *4321
# **321
# ***21
# ****1

n = 5

for i in range(n, 0, -1):
    for j in range(n, 0, -1):
        if j > i:
            print("*", end="")
        else:
            print(j, end="")
    print()
```

Pattern 20

```
# A****
# BB***
# CCC**
# DDDD*
# EEEEE

n = 5

for i in range(n):
    for j in range(n):
        if j > i:
            print("*", end="")
        else:
            print(chr(i + 65), end="")
    print()
```

Pattern 21

```
# A****
# AB***
# ABC**
# ABCD*
# ABCDE

n = 5

for i in range(n):
    for j in range(n):
        if j > i:
            print("*", end="")
        else:
            print(chr(j + 65), end="")
    print()
```

Pattern 22

```
# EEEEE
# DDDD*
# CCC**
# BB***
# A****

n = 4

for i in range(n, -1, -1):
    for j in range(n + 1):
        if j > i:
            print("*", end="")
        else:
            print(chr(i + 65), end="")
    print()
```

Pattern 23

```
# ABCDE
# ABCD*
# ABC**
# AB***
# A****

n = 4

for i in range(n, -1, -1):
    for j in range(n + 1):
        if j > i:
            print("*", end="")
        else:
            print(chr(j + 65), end="")
    print()
```

Pattern 24

```
# ****A
# ***BB
# **CCC
# *DDDD
# EEEEE

n = 4

for i in range(n + 1):
    for j in range(n, -1, -1):
        if j > i:
            print("*", end="")
        else:
            print(chr(i + 65), end="")
    print()
```

Pattern 25

```
# ****A
# ***BA
# **CBA
# *DCBA
# EDCBA


n = 4

for i in range(n + 1):
    for j in range(n, -1, -1):
        if j > i:
            print("*", end="")
        else:
            print(chr(j + 65), end="")
    print()
```

Pattern 26

```
# EEEEE
# *DDDD
# **CCC
# ***BB
# ****A

n = 4

for i in range(n, -1, -1):
    for j in range(n, -1, -1):
        if j > i:
            print("*", end="")
        else:
            print(chr(i + 65), end="")
    print()
```

Pattern 27

```
# EDCBA
# *DCBA
# **CBA
# ***BA
# ****A

n = 4

for i in range(n, -1, -1):
    for j in range(n, -1, -1):
        if j > i:
            print("*", end="")
        else:
            print(chr(j + 65), end="")
    print()
```

Pattern 28

```
# 1 2 3 4 *
# 1 2 3 * 5
# 1 2 * 4 5
# 1 * 3 4 5
# * 2 3 4 5

n = 5

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if j == n - i + 1:
            print("* ", end="")
        else:
            print(j, end=" ")
    print()
```

Pattern 29

```
#  * 1 * 2 *
#  3 * 4 * 5
#  * 6 * 7 *
#  8 * 9 *10
#  *11 *12 *

n = 5
x = 1

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if (i + j) % 2 == 1:
            print(f"{x:2}", end="")
            x += 1
        else:
            print(" *", end="")
    print()
```

Pattern 30

```
# * A* B*
# C* D* E
# * F* G*
# H* I* J
# * K* L*

n = 5
x = ord('A')

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if (i + j) % 2 == 1:
            print(chr(x), end="")
            x += 1
        else:
            print("* ", end="")
    print()
```

Pattern 31

```
# * # # # # # #
# # # # # # * *
# * * * # # # #
# # # # * * * *
# * * * # # # #
# # # # # # * *
# * # # # # # #
```

```
n = 7
x = 1

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if i % 2 == 1 and j <= x:
            print("* ", end="")
        elif i % 2 == 0 and j >= n - x + 1:
            print("* ", end="")
        else:
            print("# ", end="")
    print()
    if i <= n // 2:
        x += 1
    else:
        x -= 1
```

Pattern 32

```
# * 0 0 0 * 0 0 0 *
# 0 * 0 0 * 0 0 * 0
# 0 0 * 0 * 0 * 0 0
# 0 0 0 * * * 0 0 0
# 0 0 0 0 * 0 0 0 0
```

```
n = 5

for i in range(1, n + 1):
    for j in range(1, 2 * n):
        if j == i or j == 2 * n - i or j == n:
            print("* ", end="")
        else:
            print("0 ", end="")
    print()
```

Pattern 33

```
# 1
# 1* 2
# 1* 2* 3
# 1* 2* 3* 4
# 1* 2* 3* 4* 5

n = 5

for i in range(1, n + 1):
    for j in range(1, i + 1):
        print(j, end="")
        if j != i:
            print("* ", end="")
    print()
```

Pattern 34

```
#  1
#  3*  2
#  4*  5*  6
# 10*  9*  8*  7
# 11* 12* 13* 14* 15

n = 5

px = 1
py = 1

for i in range(1, n + 1):
    t = py
    for j in range(1, i + 1):
        if i % 2 == 1:
            print(f"{px:2}", end="")
        else:
            print(f"{t:2}", end="")
            t -= 1
        px += 1
        if j != i:
            print("* ", end="")
    py = py + i + 1
    print()
```

Pattern 35

```
# #
# **
# ###
# ****
# #####
```

```
n = 5

for i in range(1, n + 1):
    for j in range(1, i + 1):
        if i % 2 == 0:
            print("*", end="")
        else:
            print("#", end="")
    print()
```

Pattern 36

```
# 1
# **
# 333
# ****
# 55555

n = 5

for i in range(1, n + 1):
    for j in range(1, i + 1):
        if i % 2 == 0:
            print("*", end="")
        else:
            print(i, end="")
    print()
```

Pattern 37

```
# 1
# **
# 123
# ****
# 12345

n = 5

for i in range(1, n + 1):
    for j in range(1, i + 1):
        if i % 2 == 0:
            print("*", end="")
        else:
            print(j, end="")
    print()
```

Pattern 38

```
# 1
# 1*
# 1*3
# 1*3*
# 1*3*5

n = 5

for i in range(1, n + 1):
    for j in range(1, i + 1):
        if j % 2 == 0:
            print("*", end="")
        else:
            print(j, end="")
    print()
```

Pattern 39

```
# 1
# 2*
# 3*3
# 4*4*
# 5*5*5

n = 5

for i in range(1, n + 1):
    for j in range(1, i + 1):
        if j % 2 == 0:
            print("*", end="")
        else:
            print(i, end="")
    print()
```

Pattern 40

```
# #####
# ****
# ###
# **
# #
```

```
n = 5

for i in range(n, 0, -1):
    for j in range(1, i + 1):
        if i % 2 == 0:
            print("*", end="")
        else:
            print("#", end="")
    print()
```

Pattern 41

```
# #*#*#
# #*#*
# #*#
# #*
# #
```

```
n = 5

for i in range(n, 0, -1):
    for j in range(1, i + 1):
        if j % 2 == 0:
            print("*", end="")
        else:
            print("#", end="")
    print()
```

Pattern 42

```
# 55555
# ****
# 333
# **
# 1

n = 5

for i in range(n, 0, -1):
    for j in range(1, i + 1):
        if i % 2 == 0:
            print("*", end="")
        else:
            print(i, end="")
    print()
```

Pattern 43

```
# 12345
# ****
# 123
# **
# 1

n = 5

for i in range(n, 0, -1):
    for j in range(1, i + 1):
        if i % 2 == 0:
            print("*", end="")
        else:
            print(j, end="")
    print()
```

Pattern 44

```
# 1*3*5
# 1*3*
# 1*3
# 1*
# 1

n = 5

for i in range(n, 0, -1):
    for j in range(1, i + 1):
        if j % 2 == 0:
            print("*", end="")
        else:
            print(j, end="")
    print()
```

Pattern 45

```
# 1*3*5
# 1*3*
# 1*3
# 1*
# 1

n = 5

for i in range(n, 0, -1):
    for j in range(1, i + 1):
        if j % 2 == 0:
            print("*", end="")
        else:
            print(i, end="")
    print()
```

Pattern 46

```
#    #
#   **
#  ###
# ****
# #####
```

```
n = 5

for i in range(n, 0, -1):
    for j in range(1, n + 1):
        if i <= j:
            if i % 2 == 0:
                print("*", end="")
            else:
                print("#", end="")
        else:
            print(" ", end="")
    print()
```

Pattern 47

```
#     #
#    *#
#   #*#
#  *#*#
# #*#*#
```

```
n = 5

for i in range(1, n + 1):
    for j in range(n, 0, -1):
        if i >= j:
            if j % 2 == 0:
                print("*", end="")
            else:
                print("#", end="")
        else:
            print(" ", end="")
    print()
```

Pattern 48

```
#     1
#    **
#   333
#  ****
# 55555

n = 5

for i in range(1, n + 1):
    for j in range(n, 0, -1):
        if i >= j:
            if i % 2 == 0:
                print("*", end="")
            else:
                print(i, end="")
        else:
            print(" ", end="")
    print()
```

Pattern 49

```
#     1
#    **
#   321
#  ****
# 54321

n = 5

for i in range(1, n + 1):
    for j in range(n, 0, -1):
        if i >= j:
            if i % 2 == 0:
                print("*", end="")
            else:
                print(j, end="")
        else:
            print(" ", end="")
    print()
```

Pattern 50

```
#     5
#    **
#   345
#  ****
# 12345

n = 5

for i in range(n, 0, -1):
    for j in range(1, n + 1):
        if i <= j:
            if i % 2 == 0:
                print("*", end="")
            else:
                print(j, end="")
        else:
            print(" ", end="")
    print()
```

Pattern 51

```
#     1
#    *1
#   3*1
#  *3*1
# 5*3*1

n = 5

for i in range(1, n + 1):
    for j in range(n, 0, -1):
        if i >= j:
            if j % 2 == 0:
                print("*", end="")
            else:
                print(j, end="")
        else:
            print(" ", end="")
    print()
```

Pattern 52

```
#     1
#    *2
#   3*3
#  *4*4
# 5*5*5

n = 5

for i in range(1, n + 1):
    for j in range(n, 0, -1):
        if i >= j:
            if j % 2 == 0:
                print("*", end="")
            else:
                print(i, end="")
        else:
            print(" ", end="")
    print()
```

Pattern 53

```
# #####
#  ****
#   ###
#    **
#     #

n = 5

for i in range(n, 0, -1):
    for j in range(n, 0, -1):
        if i >= j:
            if i % 2 == 0:
                print("*", end="")
            else:
                print("#", end="")
        else:
            print(" ", end="")
    print()
```

Pattern 54

```
# #*#*#
#  *#*#
#   #*#
#    *#
#     #

n = 5

for i in range(n, 0, -1):
    for j in range(n, 0, -1):
        if i >= j:
            if j % 2 == 0:
                print("*", end="")
            else:
                print("#", end="")
        else:
            print(" ", end="")
    print()
```

Pattern 55

```
# 55555
#  ****
#   333
#    **
#     1

n = 5

for i in range(n, 0, -1):
    for j in range(n, 0, -1):
        if i >= j:
            if i % 2 == 0:
                print("*", end="")
            else:
                print(i, end="")
        else:
            print(" ", end="")
    print()
```

Pattern 56

```
# 54321
#  ****
#   321
#    **
#     1

n = 5

for i in range(n, 0, -1):
    for j in range(n, 0, -1):
        if i >= j:
            if i % 2 == 0:
                print("*", end="")
            else:
                print(j, end="")
        else:
            print(" ", end="")
    print()
```

Pattern 57

```
# 5*3*1
#  *3*1
#   3*1
#    *1
#     1

n = 5

for i in range(n, 0, -1):
    for j in range(n, 0, -1):
        if i >= j:
            if j % 2 == 0:
                print("*", end="")
            else:
                print(j, end="")
        else:
            print(" ", end="")
    print()
```

Pattern 58

```
#     #
#    * *
#   # # #
#  * * * *
# # # # # #
```

```
n = 5

for i in range(n, 0, -1):
    for j in range(1, n + 1):
        if i <= j:
            if i % 2 == 0:
                print("* ", end="")
            else:
                print("# ", end="")
        else:
            print(" ", end="")
    print()
```

Pattern 59

```
#     #
#    * #
#   # * #
#  * # * #
```

```
n = 5

for i in range(1, n + 1):
    for j in range(n, 0, -1):
        if i > j:
            if j % 2 == 0:
                print("* ", end="")
            else:
                print("# ", end="")
        else:
            print(" ", end="")
    print()
```

Pattern 60

```
#     1
#    * *
#   3 3 3
#  * * * *
# 5 5 5 5 5

n = 5

for i in range(1, n + 1):
    for j in range(n, 0, -1):
        if i >= j:
            if i % 2 == 0:
                print("* ", end="")
            else:
                print(f"{i}  ", end="")
        else:
            print(" ", end="")
    print()
```

Pattern 61

```
# # # # # #
#  * * * *
#   # # #
#    * *
#     #
```

```
n = 5

for i in range(n, 0, -1):
    for j in range(n, 0, -1):
        if i >= j:
            if i % 2 == 0:
                print("* ", end="")
            else:
                print("# ", end="")
        else:
            print(" ", end="")
    print()
```

Pattern 62

```
# # * # * #
#  * # * #
#   # * #
#    * #
#     #
```

```
n = 5

for i in range(n, 0, -1):
    for j in range(n, 0, -1):
        if i >= j:
            if j % 2 == 0:
                print("* ", end="")
            else:
                print("# ", end="")
        else:
            print(" ", end="")
    print()
```

Pattern 63

```
# 5 5 5 5 5
#  * * * *
#   3 3 3
#    * *
#     1

n = 5

for i in range(n, 0, -1):
    for j in range(n, 0, -1):
        if i >= j:
            if i % 2 == 0:
                print("* ", end="")
            else:
                print(f"{i} ", end="")
        else:
            print(" ", end="")
    print()
```

Pattern 64

```
# 0
# 1*
# 2**
# 3***
# 4****
# 5*****

n = 5

for i in range(n + 1):
    for j in range(i + 1):
        if j != 0:
            print("*", end="")
        else:
            print(i, end="")
    print()
```

Pattern 65

```
# 5*****
# 4****
# 3***
# 2**
# 1*
# 0

n = 5

for i in range(n, -1, -1):
    for j in range(i + 1):
        if j != 0:
            print("*", end="")
        else:
            print(i, end="")
    print()
```

Pattern 66

```
# 0
# *1
# **2
# ***3
# ****4
# *****5

n = 5

for i in range(n + 1):
    for j in range(i + 1):
        if i > j:
            print("*", end="")
        else:
            print(i, end="")
    print()
```

Pattern 67

```
# *****5
# ****4
# ***3
# **2
# *1
# 0

n = 5

for i in range(n, -1, -1):
    for j in range(i + 1):
        if i > j:
            print("*", end="")
        else:
            print(i, end="")
    print()
```

Pattern 68

```
# *   *
#  * *
#   *
#  * *
# *   *
```

```
n = 5

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if i == j or i + j == n + 1:
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

Pattern 69

```
#   *
#   *
# *****
#   *
#   *
```

```
for i in range(1, 6):
    for j in range(1, 6):
        if j == 3 or i == 3:
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

Pattern 70

```
# *****
# *   *
# *   *
# *   *
# *****

n = 5

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if j == 1 or j == n or i == 1 or i == n:
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

Pattern 71

```
# *****
# *###*
# *###*
# *###*
# *****

n = 5

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if j == 1 or j == n or i == 1 or i == n:
            print("*", end="")
        else:
            print("#", end="")
    print()
```

Pattern 72

```
# *******
# **   **
# * * * *
# *  *  *
# * * * *
# **   **
# *******

n = 7

for i in range(n):
    for j in range(n):
        if i == 0 or j == 0 or i == j or i == n - 1 or j == n - 1 or i + j == n - 1:
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

Pattern 73

```
# *
# * *
# *   *
# *     *
# *       *
# *         *
# * * * * * * *
```

```
n = 7

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if j == 1 or i == j or i == n:
            print("* ", end="")
        else:
            print("  ", end="")
    print()
```

Pattern 74

```
# * * * * * * * *
# *         *
# *       *
# *     *
# *   *
# * *
# *
```

```
n = 7
for i in range(n, 0, -1):
    for j in range(n, 0, -1):
        if j == n - i + 1 or j == n or i == n:
            print("* ", end="")
        else:
            print("  ", end="")
    print()
```

Pattern 75

```
#        *
#      * *
#    *   *
#  *     *
# * * * * *
```

```
n = 5
for i in range(1, n + 1):
    for j in range(1, n + 1):
        if j == n - i + 1 or j == n or i == n:
            print("* ", end="")
        else:
            print("  ", end="")
    print()
```

Pattern 76

```
# * * * * *
#  *    *
#   *  *
#    * *
#      *
```

```
n = 5
for i in range(n, 0, -1):
    for j in range(1, n + 1):
        if j == n - i + 1 or j == n or i == n:
            print("* ", end="")
        else:
            print("  ", end="")
    print()
```

Pattern 77

```
#     *
#    * *
#   *   *
#  *     *
# *       *
```

```
n = 5
px = n
py = n

for i in range(1, n + 1):
    for j in range(1, n * 2 + 1):
        if j == px or j == py:
            print("*", end="")
        else:
            print(" ", end="")
    px -= 1
    py += 1
    print()
```

Pattern 78

```
# *      *
#  *    *
#   *  *
#    * *
#     *
```

```
n = 5
px = 1
py = n * 2 - 1

for i in range(1, n + 1):
    for j in range(1, n * 2 + 1):
        if j == px or j == py:
            print("*", end="")
        else:
            print(" ", end="")
    px += 1
    py -= 1
    print()
```

Pattern 79

```
# *
#  *
#   *
#    *
#   *
#  *
# *
```

```
n = 7
x = 1
for i in range(1, n + 1):
    for j in range(1, x + 1):
        if j == x:
            print("* ", end="")
        else:
            print(" ", end="")
    print()
    if i <= n / 2:
        x += 1
    else:
        x -= 1
```

Pattern 80

```
#    *
#   *
#  *
# *
#  *
#   *
#    *
```

```
n = 7
x = n // 2 + 1

for i in range(1, n + 1):
    for j in range(1, x + 1):
        if j == x:
            print("* ", end="")
        else:
            print("  ", end="")
    print()

    if i <= n // 2:
        x -= 1
    else:
        x += 1
```

Pattern 81

```
#    *
#   * *
#  *   *
# *     *
#*       *
# *     *
#  *   *
#   * *
#    *
```

```
n = 9
px = n // 2 + 1

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if j == px or j == n - px + 1:
            print("*", end="")
        else:
            print(" ", end="")
    if i <= n // 2:
        px -= 1
    else:
        px += 1
    print()
```

Pattern 82

```
#     *
#    / \
#   /   \
#  /     \
# /       \
#\       /
# \     /
#  \   /
#   \ /
#    *
```

```
n = 10

px = n // 2
py = n // 2

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if (i == 1 or i == n) and j == n // 2:
            print("*", end="")
        elif j == px:
            print("/", end="")
        elif j == py:
            print("\\", end="")
        else:
            print(" ", end="")

    if i == n // 2:
        px, py = py, px
    else:
        px -= 1
        py += 1
    print()
```

Pattern 83

```
#     *
#    - -
```

```
#   -   -
#  -     -
# -       -
# -       -
#  -     -
#   -   -
#    - -
#     *

n = 10
px = n // 2
py = n // 2

for i in range(1, n + 1):
    for j in range(1, n):
        if (i == 1 or i == n) and j == n // 2:
            print("*", end="")
        elif j == px or j == py:
            print("-", end="")
        else:
            print(" ", end="")
    if i == n // 2:
        t = px
        px = py
        py = t
    else:
        px -= 1
        py += 1
    print()
```

Pattern 84

```
#     *
#    * *
#   * * *
#  * * * *
#   * * *
#    * *
```

```
#    *

size = 3

for i in range(size, -size-1, -1):
    for j in range(1, abs(i)+1):
        print(" ", end="")
    for k in range(size, abs(i)-1, -1):
        print("* ", end="")
    print()
```

Pattern 85

```
# *
# **
# ***
# *
# **
# ***
# *
# **
# ***

n = 3
for i in range(3):
    for j in range(1, n+1):
        print("*" * j)
```

Pattern 86

```
# *
# * *
# 123
# * *
# *
```

```
#include <stdio.h>
```

```
n = 5
x = 1

for i in range(1, n+1):
    for j in range(1, x+1):
        if i != n//2 + 1:
            print("* ", end="")
        else:
            print(j, end=" ")
    if i <= n//2:
        x += 1
    else:
        x -= 1
    print()
```

Pattern 87

```
# *
# * *
# ABC
# * *
# *

n = 5
x = 1

for i in range(1, n + 1):
    for j in range(1, x + 1):
        if i != n // 2 + 1:
            print("*", end=" ")
        else:
            print(chr(64 + j), end=" ")
    if i <= n // 2:
        x += 1
    else:
        x -= 1
    print()
```

Pattern 88

```
# 1
# 2* 2
# 3* 3* 3
# 4* 4* 4* 4
# 3* 3* 3
# 2* 2
# 1

n = 7
x = 1

for i in range(1, n + 1):
    for j in range(1, 2 * x):
        if i <= n // 2 + 1 and j % 2 == 1:
            print(i, end="")
        elif i > n // 2 + 1 and j % 2 == 1:
            print(n - i + 1, end="")
        else:
            print("*", end=" ")
    if i <= n // 2:
        x += 1
    else:
        x -= 1
    print()
```

Pattern 89

```
# * * * * * * *
#   * * * * *
#     * * *
#       *
#     * * *
#   * * * * *
# * * * * * * *

n = 7
x = 1

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if j >= x and j <= n - x + 1:
            print("* ", end="")
        else:
            print("  ", end="")
    if i <= n // 2:
        x += 1
    else:
        x -= 1
    print()
```

Pattern 90

```
# *         *
# * *     * *
# * * *   * * *
# * * * * * * *
# * * *   * * *
# * *     * *
# *         *
```

```
n = 7
x = 1

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if j <= x or j >= n - x + 1:
            print("* ", end="")
        else:
            print("  ", end="")
    if i <= n // 2:
        x += 1
    else:
        x -= 1
    print()
```

Pattern 91

```
# * * * * *
# * * * *
# * * *
# * *
# *
# * *
# * * *
# * * * *
# * * * * *
```

```
size = 4

for i in range(size, -size - 1, -1):
    for j in range(0, abs(i) + 1):
        print("* ", end="")
    print()
```

Pattern 92

```
# * * * *
#   * * *
#     * *
#       *
#     * *
#   * * *
# * * * *

n = 7
x = 1

for i in range(1, n + 1):
    for j in range(1, n // 2 + 2):
        if j >= x:
            print("* ", end="")
        else:
            print("  ", end="")
    if i <= n // 2:
        x += 1
    else:
        x -= 1
    print()
```

Pattern 93

```
# *       *
# **     **
# ***   ***
# **** ****
# *********
```

```
n = 5

for i in range(1, n + 1):
    for j in range(1, n * 2):
        if j <= i or j >= n * 2 - i:
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

Pattern 94

```
# *********
# **** ****
# ***   ***
# **     **
# *       *
```

```
n = 5
px = n
py = n

for i in range(1, n + 1):
    for j in range(1, n * 2):
        if j > px and j < py:
            print(" ", end="")
        else:
            print("*", end="")
    px -= 1
    py += 1
    print()
```

Pattern 95

```
#    *       *
#   ***     ***
#  *****   *****
# *******  *******
# ******************
```

```
min_stars = 1

p_height = 5
p_space = p_height - 1

for i in range(p_height):
    for j in range(p_space, i, -1):
        print(" ", end="")
    for k in range(min_stars):
        print("*", end="")
    for l in range(p_space * 2, p_space - 1, -1):
        print(" ", end="")
    for m in range(min_stars):
        print("*", end="")

    min_stars += 2
    print()
```

Pattern 96

```
# 1 2 3 4 * 4 3 2 1
# 1 2 3 * * * 3 2 1
# 1 2 * * * * * 2 1
# 1 * * * * * * * 1
# * * * * * * * * *

n = 5

x = n

for i in range(1, n + 1):
    for j in range(1, 2 * n):
        if j >= x and j <= n + i - 1:
            print("* ", end="")
        elif j <= n:
            print(j, end=" ")
        else:
            print(2 * n - j, end=" ")
    x -= 1
    print()
```

Pattern 97

```
#         *
#       # * #
#     * # * # *
#   # * # * # * #
# * # * # * # * # *
```

```
n = 5

x = n

for i in range(1, n + 1):
    for j in range(1, 2 * n):
        if j >= x and j < n + i and j % 2 == 1:
            print("* ", end="")
        elif j >= x and j < n + i and j % 2 == 0:
            print("# ", end="")
        else:
            print("  ", end="")  # Adjust spacing for better alignment
    x -= 1
    print()
```

Pattern 98

```
#         *
#        * *
#       * * *
#      * * * *
#     *       *
#    * *     * *
#   * * *   * * *
#  * * * * * * * * *
# * * * * * * * * * * *
```

```
p_height = 5
p_space = p_height - 1
x = 1

# Upper part of the pattern
for i in range(1, p_height):
    print(" " * p_height, end="")
    for k in range(p_space, i - 1, -1):
        print(" ", end="")
    for j in range(1, i + 1):
        print("* ", end="")
    print()

# Lower part of the pattern
for i in range(p_height):
    for j in range(p_space, i, -1):
        print(" ", end="")
    for k in range(i + 1):
        print("* ", end="")
    for n in range(p_height + p_height - 2, x - 1, -1):
        print(" ", end="")
    for k in range(i + 1):
        print("* ", end="")
    x += 2
    print()
```

Pattern 99

```
#     *
#    * *
#   *   *
#  *     *
# *********
```

```
n = 5
px = n
py = n

for i in range(1, n + 1):
    for j in range(1, n * 2):
        if j == px or j == py or i == n:
            print("*", end="")
        else:
            print(" ", end="")
    px -= 1
    py += 1
    print()
```

Pattern 100

```
# *
# **
# * *
# *  *
# * *
# **
# *
```

```
n = 7

x = 1

for i in range(1, n + 1):
    for j in range(1, x + 1):
        if j == 1 or j == x:
            print("*", end="")
        else:
            print(" ", end="")
    print()

    if i <= n // 2:
        x += 1
    else:
        x -= 1
```

Pattern 101

```
#    *
#   **
#  * *
# *  *
#  * *
#   **
#    *

n = 7
x = n // 2 + 1

for i in range(1, n + 1):
    for j in range(1, n // 2 + 2):
        if j == x or j == n // 2 + 1:
            print("*", end="")
        else:
            print(" ", end="")
    if i <= n // 2:
        x -= 1
    else:
        x += 1
    print()
```

Pattern 102

```
# *********
#  *     *
#   *   *
#    * *
#     *
```

```
n = 5
x = 1
y = 2 * n - 1

for i in range(1, n + 1):
    for j in range(1, 2 * n):
        if j == x or j == y or i == 1:
            print("*", end="")
        else:
            print(" ", end="")
    print()
    x += 1
    y -= 1
```

Pattern 103

```
# *       *
# * *     * *
# *  *  *  *
# *    *    *
# *  *  *  *
# * *     * *
# *       *
```

```
n = 7
x = 1

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if j == x or j == n - x + 1 or (j == 1 and i != 1 and i != n) or (j == n and i
!= 1 and i != n):
            print("* ", end="")
        else:
            print("  ", end="")
    if i <= n // 2:
        x += 1
    else:
        x -= 1
    print()
```

Pattern 104

```
# *   *   *
#  *  *  *
#   * * *
# * * * * * * *
#   * * *
#  *  *  *
# *   *   *
```

```
n = 7

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if j == i or j == n - i + 1 or i == n // 2 + 1 or j == n // 2 + 1:
            print("* ", end="")
        else:
            print("  ", end="")
    print()
```

Pattern 105

```
# *    * * * *
# *    *
# *    *
# * * * * * * *
#      *    *
#      *    *
# * * * *    *
```

```
n = 7

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if i == n // 2 + 1 or j == n // 2 + 1 or \
           (i == 1 and j >= n // 2 + 1) or \
           (i == n and j <= n // 2) or \
           (j == 1 and i <= n // 2) or \
           (j == n and i >= n // 2 + 1):
            print("* ", end="")
        else:
            print("  ", end="")
    print()
```

Pattern 106

```
# *   *   *   *
# *   *   *   *
# * * * * * * *
# *           *
# *           *
# *           *
# * * * * * * *
```

```
n = 7

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if (i == n // 2 or i == n) or \
           ((j == 1 or j == n) and i >= n // 2) or \
           (j % 2 == 1 and i <= n // 2):
            print("* ", end="")
        else:
            print("  ", end="")
    print()
```

Pattern 107

```
# * * * * * * * * *
# *           *
# *   * * * * *   *
# *   *       *   *
# *   *   *   *   *
# *   *       *   *
# *   * * * * *   *
# *           *
# * * * * * * * * *

n = 9
x = 1

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if (j <= x or j >= n - x + 1) and j % 2 == 1:
            print("* ", end="")
        elif (j >= x and j <= n - x + 1) and i % 2 == 1:
            print("* ", end="")
        else:
            print("  ", end="")
    print()
    if i <= n // 2:
        x += 1
    else:
        x -= 1
```

Pattern 108

```
#  *  *  *
# *  *  *  *
#  *  *  *
# *  *  *  *
#  *  *  *
# *  *  *  *
#  *  *  *
```

```
n = 7

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if (i + j) % 2 == 1:
            print("* ", end="")
        else:
            print("  ", end="")
    print()
```

Pattern 109

```
# *     *     *
# **   ***   **
# *** ***** ***
# ***********
# ***********
# ***********
# ***********
```

```
n = 7

x = n
for i in range(1, n + 1):
    for j in range(1, 2 * n):
        if (j >= x and j <= 2 * n - x) or j <= n - x + 1 or j >= n + x - 1:
            print("*", end="")
        else:
            print(" ", end="")
    x -= 1
    print()
```

Pattern 110

```
#       *
#     * * *
#   *   *   *
# *     *     *
# * * * * * * * * *
# *     *     *
#   *   *   *
#     * * *
#       *
```

```
n = 9
c1 = (n - 1) // 2
c2 = 3 * n // 2 - 1

for i in range(n):
    for j in range(n):
        if i + j == c1 or i - j == c1 or j - i == c1 or i + j == c2 or i == c1 or j ==
c1:
            print("* ", end="")
        else:
            print("  ", end="")
    print()
```

Pattern 111

```
# * * * * * * * * *
# * * * *   * * * *
# * * *       * * *
# * *           * *
# *               *
# * *           * *
# * * *       * * *
# * * * *   * * * *
# * * * * * * * * *
```

```
n = 9
c1 = (n - 1) // 2
c2 = 3 * n // 2 - 1

for i in range(n):
    for j in range(n):
        if i + j <= c1 or i - j >= c1 or j - i >= c1 or i + j >= c2:
            print("* ", end="")
        else:
            print("  ", end="")
    print()
```

Pattern 112

```
# * * * * * * * * *
# * * *   * * * *
# * *       * * *
# *           * *
# * *           *
# * * *       * *
# * * * *   * * *
# * * * * * * * * *
```

```
n = 8
c1 = (n - 1) // 2
c2 = 3 * n // 2 - 1

for i in range(n):
    for j in range(n):
        if i + j <= c1 or i - j >= c1 or j - i >= c1 or i + j >= c2:
            print("* ", end="")
        else:
            print("  ", end="")
    print()
```

Pattern 113

```
# *     *
# **   **
# *** ***
# *******
# *** ***
# **   **
# *     *
```

```
n = 7
px = 1

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if j <= px or j >= n - px + 1:
            print("*", end="")
        else:
            print(" ", end="")

    if i <= n // 2:
        px += 1
    else:
        px -= 1

    print()
```

Pattern 114

```
#     *****
#    *****
#   *****
#  *****
# *****
```

```
z = 1
n = 5

for i in range(n):
    for j in range(n - 1, i, -1):
        print(" ", end="")
    for k in range(n):
        print("*", end="")
    z += 1
    print()
```

Pattern 115

```
# **
# **
# ****
# ****
# ******
# ******

for i in range(1, 7):
    k = i + 1 if i % 2 != 0 else i
    print("*" * k)
```

Pattern 116

```
#    **
#    **
#  ****
#  ****
# ******
# ******
```

```
for i in range(1, 7):
    k = i + 1 if i % 2 != 0 else i
    print(" " * (6 - k), "*" * k)
```

Pattern 117

```
# ******
# ******
# ****
# ****
# **
# **
```

```
n = 6

for i in range(n, 0, -1):
    k = i + 1 if i % 2 != 0 else i
    print("*" * k)
```

Pattern 118

```
# ******
# ******
#   ****
#   ****
#     **
#     **
```

```
n = 6

for i in range(n, 0, -1):
    k = i + 1 if i % 2 != 0 else i
    print(" " * (n - k) + "*" * k)
```

Pattern 119

```
# *
# *
# **
# **
# ***
# ***
```

```
n = 3

for i in range(1, n + 1):
    for c in range(n - 1, 0, -1):
        for j in range(i):
            print("*", end="")
        print()
```

Pattern 120

```
#   *
#   *
#  **
#  **
# ***
# ***
```

```
for i in range(1, 4):
    for x in range(2, 0, -1):
        for y in range(2, i - 1, -1):
            print(" ", end="")
        for j in range(i):
            print("*", end="")
        print()
```

Pattern 121

```
# ***
# ***
# **
# **
# *
# *
```

```
for i in range(3, 0, -1):
    for x in range(2, 0, -1):
        for j in range(i):
            print("*", end="")
        print()
```

Pattern 122

```
# ***
# ***
#  **
#  **
#   *
#   *
```

```
for i in range(3, 0, -1):
    for x in range(2, 0, -1):
        for y in range(2, i - 1, -1):
            print(" ", end="")
        for j in range(i):
            print("*", end="")
        print()
```

Pattern 123

```
#   *
#   *
#  ***
#  ***
# *****
# *****
```

```
n = 3
z = 1
for i in range(n):
    for j in range(2, 0, -1):
        for x in range(n - 1, i - 1, -1):
            print(" ", end="")
        for y in range(z):
            print("*", end="")
        print()
    z += 2
```

Pattern 124

```
#      **
#      **
#    *   *
#    *   *
#  *       *
#  *       *
# *         *

n = 7
x = n
for i in range(1, n + 1):
    for j in range(1, 2 * n + 1):
        if j == x or j == 2 * n - x + 1:
            print("*", end="")
        else:
            print(" ", end="")
    print()
    if i % 2 == 0:
        x -= 2
```

Pattern 125

```
# *****
#  ***
#   *
#  ***
# *****

n = 3
s1 = n * 2 - 1
s2 = 3

for i in range(1, n + 1):
    print("  " * (i - 1), end="")
    print("*" * s1)
    s1 -= 2

for i in range(1, n):
    print("  " * (n - i - 1), end="")
    print("*" * s2)
    s2 += 2
```

Pattern 126

```
#    * * * * * *
#   * * * * * * * *
#  * * * * * * * * *
# * * * * * * * * * *
# * * * * * * * * * *
# * * * * * * * * * *
#  * * * * * * * * *
#   * * * * * * * *
#    * * * * * *
```

```
for i in range(1, 10):
    if i == 1 or i == 9:
        x = 6
        z = 4
    elif i == 2 or i == 8:
        x = 8
        z = 2
    elif i == 3 or i == 7:
        x = 9
        z = 1
    else:
        x = 10
        z = 0

    print(" " * z, end="")
    print("* " * x)
```

Pattern 127

```
#  ***   ***
# ***** *****
# ***********
#  *********
#   *******
#    *****
#     ***
#      *
```

```
n = 6

# Top part of the heart
for i in range(n // 2, n + 1, 2):
    print(" " * (n - i), end="")
    print("*" * i, end="")
    print(" " * (n - i), end="")
    print("*" * i)

# Bottom part of the heart
for i in range(n, 0, -1):
    print(" " * (n - i), end="")
    print("*" * (2 * i - 1))
```

Pattern 128

```
# ** **
# *  *  *
# *     *
#  *   *
#   * *
#    *

for row in range(6):
    for col in range(7):
        if (row == 0 and col % 3 != 0) or (row == 1 and col % 3 == 0) or (row
- col == 2) or (row + col == 8):
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

Pattern 129: X-Max Tree

```
#        *
#       * *
#      * * *
#     * * * *
#      * * *
#     * * * *
#    * * * * *
#   * * * * * *
#     * * * *
#     * * * *
#     * * * *
#     * * * *
```

```
height = 3
width = 4

space = width * 5
r = 1
m = 1

for r in range(1, height):
    for i in range(m, width + 1):
        for j in range(space, i - 1, -1):
            print(" ", end="")
        for k in range(1, i + 1):
            print("* ", end="")
        print()
    m += 2
    width += 2

for i in range(1, 5):
    for j in range(space - 3, 0, -1):
        print(" ", end="")
    for k in range(1, 5):
        print("* ", end="")
    print()
```

Pattern 130

```
# * * * * *
# *   * *
# *  *  *
# * *   *
# * * * * *
```

```
n = 5

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if i == 1 or i == n or j == 1 or j == n or i + j == n + 1:
            print("* ", end="")
        else:
            print("  ", end="")
    print()
```

Pattern 131

```
#   * *
# *   *
#     *
#     *
# * * * * *
```

```
n = 5
mid = n // 2 + 1

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if j == mid or i == n or (i + j == mid and j <= mid):
            print("* ", end="")
        else:
            print("  ", end="")
    print()
```

Pattern 132

```
# * * * * *
#       *
# * * * * *
# *
# * * * * *
```

```
n = 5
mid = n // 2 + 1

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if i == 1 or i == n or i == mid or (j == 1 and i >= mid) or (j == n and i <
mid):
            print("* ", end="")
        else:
            print("  ", end="")
    print()
```

Pattern 133

```
# * * * * *
#       *
# * * * * *
#       *
# * * * * *
```

```
n = 5
mid = n // 2 + 1

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if i == 1 or i == n or i == mid or j == n:
            print("* ", end="")
        else:
            print("  ", end="")
    print()
```

Pattern 134

```
# *     *
# *     *
# * * * * *
#       *
#       *
```

```
n = 5
mid = n // 2 + 1

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if i == mid or (j == 1 and i <= mid) or j == n:
            print("* ", end="")
        else:
            print("  ", end="")
    print()
```

Pattern 135

```
# * * * * *
# *
# * * * * *
#       *
# * * * * *
```

```
n = 5
mid = n // 2 + 1

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if i == 1 or i == n or i == mid or (j == 1 and i <= mid) or (j == n and i >
mid):
            print("* ", end="")
        else:
            print("  ", end="")
    print()
```

Pattern 136

```
# * * * * *
# *
# * * * * *
# *       *
# * * * * *
```

```
n = 5
mid = n // 2 + 1

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if i == 1 or i == n or i == mid or j == 1 or (j == n and i > mid):
            print("* ", end="")
        else:
            print("  ", end="")
    print()
```

Pattern 137

```
# * * * * *
#       *
#     *
#   *
# *
```

```
n = 5

for i in range(1, n + 1):
    for j in range(1, n):
        if i == 1 or i + j == n + 1:
            print("* ", end="")
        else:
            print("  ", end="")
    print()
```

Pattern 138

```
# * * * * *
# *     *
# * * * * *
# *     *
# * * * * *
```

```
n = 5
mid = n // 2 + 1

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if i == 1 or i == n or i == mid or j == 1 or j == n:
            print("* ", end="")
        else:
            print("  ", end="")
    print()
```

Pattern 139

```
# * * * * *
# *       *
# * * * * *
#         *
# * * * * *
```

```
n = 5
mid = n // 2 + 1

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if i == 1 or i == n or i == mid or (j == 1 and i <= mid) or j == n:
            print("* ", end="")
        else:
            print("  ", end="")
    print()
```

Pattern 140

```
#     *
#    * *
#   *****
#  *     *
# *       *
```

```
n = 5
x = n

for i in range(1, n + 1):
    for j in range(1, 2 * n + 1):
        if j == x or j == n + i - 1:
            print("*", end="")
        elif j >= x and j <= n + i - 1 and i == n // 2 + 1:
            print("*", end="")
        else:
            print(" ", end="")
    print()
    x -= 1
```

Pattern 141

```
# ****
# *   *
# *****
# *   *
# ****

n = 5

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if j == 1 or (i == 1 or i == n or i == n // 2 + 1) and j <= n - 1:
            print("*", end="")
        elif i != 1 and i != n and j == n:
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

Pattern 142

```
# *****
# *
# *
# *
# *****

n = 5

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if i == 1 or i == n or j == 1:
            print("*", end="")
    print()
```

Pattern 143

```
# ****
# *   *
# *   *
# *   *
# ****

n = 5

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if j == 1 or (i == 1 or i == n) and j <= n - 1:
            print("*", end="")
        elif i != 1 and i != n and j == n:
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

Pattern 144

```
# *****
# *
# *****
# *
# *****
```

```
n = 5

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if j == 1 or i == 1 or i == n or i == n // 2 + 1:
            print("*", end="")
    print()
```

Pattern 145

```
# *****
# *
# *****
# *
# *
```

```
n = 5

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if j == 1 or i == 1 or i == n // 2 + 1:
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

Pattern 146

```
# *****
# *
# * ***
# *   *
# *****

n = 5

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if j == 1 or i == 1 or i == n or (i > n // 2 and j == n) or (i == (n // 2 + 1)
and j > n // 2):
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

Pattern 147

```
# *     *
# *     *
# * * * * *
# *     *
# *     *
```

```
n = 5

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if j == 1 or j == n or i == n // 2 + 1:
            print("*", end=" ")
        else:
            print(" ", end=" ")
    print()
```

Pattern 148

```
# * * * * *
#   *
#   *
#   *
# * * * * *

n = 5

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if i == 1 or i == n or j == n // 2 + 1:
            print("*", end=" ")
        else:
            print(" ", end=" ")
    print()
```

Pattern 149

```
# * * * * *
#     *
#     *
#     *
# * * *
```

```
n = 5

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if i == 1 or (i == n and j <= n // 2 + 1) or j == n // 2 + 1:
            print("*", end=" ")
        else:
            print(" ", end=" ")
    print()
```

Pattern 150

```
# *   *
# * *
# *
# * *
# *   *

n = 5

x = n // 2 + 1

for i in range(1, n + 1):
    for j in range(1, n // 2 + 2):
        if j == 1 or j == x:
            print("*", end=" ")
        else:
            print(" ", end=" ")
    if i <= n // 2:
        x -= 1
    else:
        x += 1
    print()
```

Pattern 151

```
# *
# *
# *
# *
# * * *
```

```
n = 5

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if j == 1 or (i == n and j <= (n # 2) + 1):
            print("* ", end="")
    print()
```

Pattern 152

```
# *     *
# * *  * *
# *  *  *
# *     *
# *     *
```

```
n = 5
x = 1

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if j == 1 or j == n or (j == x or j == n - x + 1) and i <= (n // 2) + 1:
            print("* ", end="")
        else:
            print("  ", end="")
    x += 1
    print()
```

Pattern 153

```
# *      *
# * *    *
# *  *   *
# *    * *
# *      *
```

```
n = 5

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if j == 1 or j == n or i == j:
            print("* ", end="")
        else:
            print("  ", end="")
    print()
```

Pattern 154

```
# * * * * *
# *       *
# *       *
# *       *
# * * * * *
```

```
n = 5

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if j == 1 or j == n or i == 1 or i == n:
            print("* ", end="")
        else:
            print("  ", end="")
    print()
```

Pattern 155

```
# * * *
# *   *
# * * *
# *
# *
```

```
n = 5
x = (n // 2) + 1

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if (i == 1 and j <= x) or j == 1 or (j == x and i <= x) or (i == x and j <=
x):
            print("* ", end="")
        else:
            print("  ", end="")
    print()
```

Pattern 156

```
# * * * * * * *
# *         *
# *         *
# *         *
# *       * *
# * * * * * * *
#             *
```

```
n = 7
x = n - n // 2 + 1

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if (i == 1 or i == x + 1) and j <= x + 1 or (j == 1 or j == x + 1) and i <=
x + 1 or (j == i and i >= x):
            print("* ", end="")
        else:
            print("  ", end="")
    print()
```

Pattern 157

```
# * * *
# *   *
# * * *
# * *
# *   *
```

```
n = 5

for i in range(1, n + 1):
    x = 1
    for j in range(1, (n // 2) + 2):
        if j == 1 or i == 1 or i == (n // 2) + 1 or (i <= n // 2 and j == (n // 2) +
1) or j == x:
            print("* ", end="")
        else:
            print("  ", end="")
        x += 1 if i > n // 2 else 0
    print()
```

Pattern 158

```
# * * * * *
# *
# * * * * *
#       *
# * * * * *
```

```
n = 5
for i in range(1, n + 1):
    for j in range(1, n + 1):
        if i == 1 or i == n or i == n // 2 + 1 or (j == 1 and i <= n // 2 + 1) or (j
== n and i > n // 2 + 1):
            print("* ", end="")
        else:
            print("  ", end="")
    print()
```

Pattern 159

```
# * * * * *
#     *
#     *
#     *
#     *

n = 5
for i in range(1, n + 1):
    for j in range(1, n + 1):
        if i == 1 or j == n // 2 + 1:
            print("* ", end="")
        else:
            print("  ", end="")
    print()
```

Pattern 160

```
# *     *
# *     *
# *     *
# *     *
# * * * * *
```

```
n = 5
for i in range(1, n + 1):
    for j in range(1, n + 1):
        if j == 1 or j == n or i == n:
            print("* ", end="")
        else:
            print("  ", end="")
    print()
```

Pattern 161

```
# *       *
#  *     *
#   *   *
#    * *
#     *

n = 5
x = 1
for i in range(1, n + 1):
    for j in range(1, 2 * n):
        if j == x or j == 2 * n - x:
            print("*", end="")
        else:
            print(" ", end="")
    print()
    x += 1
```

Pattern 162

```
# *   *   *
# *  * *  *
# * *   * *
# **    **
# *      *
```

```
n = 5
x = n

for i in range(1, n + 1):
    for j in range(1, 2 * n):
        if j == 1 or j == 2 * n - 1 or j == x or j == 2 * n - x:
            print("*", end="")
        else:
            print(" ", end="")
    print()
    x -= 1
```

Pattern 163

```
# *     *
#  *  *
#    *
#  *  *
# *     *
```

```
n = 5

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if j == i or j == n - i + 1:
            print("* ", end="")
        else:
            print("  ", end="")
    print()
```

Pattern 164

```
# *     *
#  *  *
#   *
#   *
#   *

n = 5

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if (j == i or j == n - i + 1) and i <= n // 2 + 1 or (j == n // 2 + 1 and i > n
// 2):
            print("* ", end="")
        else:
            print("  ", end="")
    print()
```

Pattern 165

```
# * * * * *
#       *
#     *
#   *
# * * * * *
```

```
n = 5

for i in range(1, n + 1):
    for j in range(1, n + 1):
        if i == 1 or i == n or j == n - i + 1:
            print("* ", end="")
        else:
            print("  ", end="")
    print()
```

In every line of code, they have woven a story of innovation and creativity. This book has been your compass in the vast world of Python.

Close this chapter knowing that every challenge overcome is an achievement, and every solution is a step toward mastery.

Your code is the melody that gives life to projects. May they continue creating and programming with passion!

Thank you for allowing me to be part of your journey.

With gratitude,
Hernando Abella
Author of 150+ Python Pattern Programs

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